TASK # 2

SOFTWARE CONSTRUCTION AND DEVELOPMENT LAB

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BSSE012315010

Home telephone system line (class diagram)

Lab Task Description

The below diagram represents a UML Class Diagram for a traditional/old-style telephone system. This diagram models the structural components and relationships found in a classic landline telephone setup.

What We Have Done:

- Analyzed the components of a traditional telephone system
- Identified the main classes, their attributes, and methods
- Established relationships (associations, multiplicities) between components
- Created a visual UML representation showing how telephone parts interconnect

What You Need to Read/Understand:

The diagram shows:

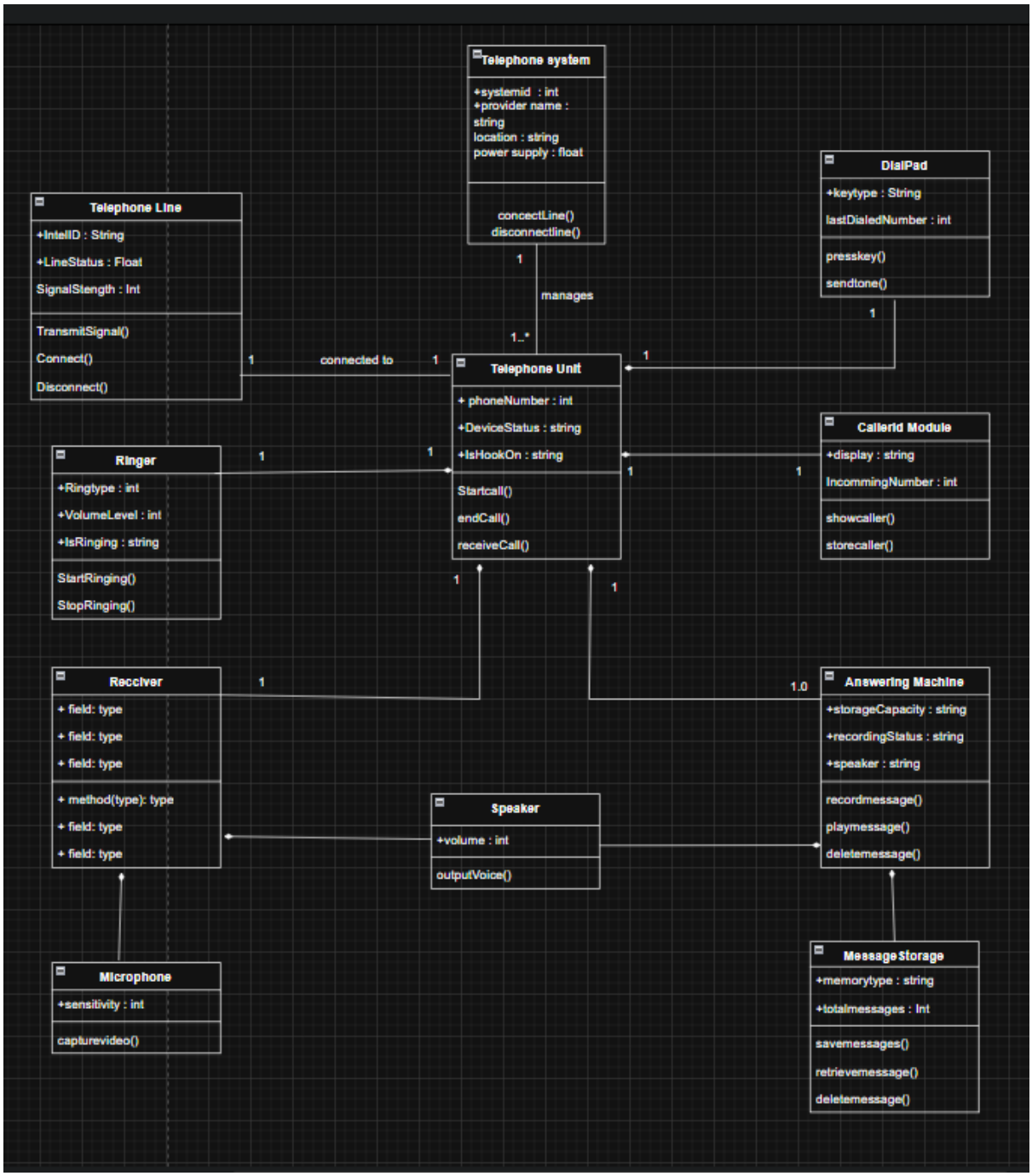
1.Telephone System(main system class) - manages the overall telephone network connection

2. Telephone Unit (core device) - represents the physical telephone handset

3. Supporting Components:

   - DialPad - for entering numbers
   - Ringer - for incoming call alerts
   - Receiver - receives audio
   - Speaker - outputs sound
   - Microphone - captures voice
   - CallerId Module - displays incoming number
   - Answering Machine - records messages
   - Message Storage - stores recorded messages

Relationships:

- One Telephone System manages 1 or more Telephone Units
- Each Telephone Unit has exactly one DialPad, Ringer, Receiver, Speaker, Microphone
- Optional components: CallerId Module (0 or 1), Answering Machine (0 or 1)
- Answering Machine connects to Message Storage (1-to-1)

Class Diagram: